Exact value equations

1 State whether the following equations have a solution:

a $\cos\theta - 4 = 0$

b $9\tan\theta + 4 = 0$

2 State the number of solutions for θ of the following equations in the domain $0^\circ < \theta < 90^\circ$.

a $\cos\theta = -\frac{1}{\sqrt{2}}$

b $\sin\theta = -\frac{\sqrt{3}}{2}$

c $\tan\theta = -1$

3 Solve the following equations for $0^\circ \leq \theta \leq 90^\circ$:

a $\sin\theta = \frac{1}{\sqrt{2}}$

b $\tan\theta = \sqrt{3}$

c $\cos\theta = \frac{1}{2}$

d $\sin\theta = \frac{\sqrt{3}}{2}$

4 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$:

a $\cos\theta = -\frac{1}{\sqrt{2}}$

b $\cos\theta = \frac{1}{2}$

c $\cos\theta = 0$

d $\sin\theta = \frac{1}{2}$

e $\sin\theta = 0$

f $\sin\theta = -\frac{1}{\sqrt{2}}$

g $\cos\theta = -\frac{1}{\sqrt{2}}$

h $\sin\theta = -\frac{\sqrt{3}}{2}$

i $\sin\theta = 1$

j $\tan\theta = \sqrt{3}$

k $\tan\theta = 0$

l $\tan\theta = -\frac{1}{\sqrt{3}}$

m $4\tan\theta + 2 = -2$

n $8\cos\theta - 4 = 0$

o $2\cos\theta + 4 = 3$

p $8\sin\theta - 4\sqrt{2} = 0$

Non-exact value equations

5 Solve the following equations for $0^\circ \leq \theta \leq 90^\circ$:

a $\cos\theta = 0.7986$ b $\sin\theta = 0.6428$ c $\tan\theta = 0.7265$ d $\sin\theta = 0.3584$

e $\tan\theta = 2.2460$

6 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$:

a $\cos\theta = 0.9063$ b $\cos\theta = -0.7986$ c $\sin\theta = -0.6428$ d $\sin\theta = 0.9336$

e $\tan\theta = 0.7002$ f $\tan\theta = -0.7265$

Different domains

7 State the number of solutions for the following equations in the domain $-90^\circ < \theta < 90^\circ$:

a $\cos\theta = -\dfrac{1}{\sqrt{2}}$ b $\cos\theta = -0.914$

8 State the number of solutions for the following equations in the domain $180^\circ \leq \theta \leq 360^\circ$:

a $\sin\theta = -\dfrac{1}{\sqrt{2}}$ b $\sin\theta = -0.697$

9 Solve the following equations for the given domain:

a $\cos\theta = \dfrac{1}{\sqrt{2}}$ for $0^\circ < \theta < 90^\circ$ b $\sin\theta = -\dfrac{1}{2}$ for $-90^\circ \leq \theta \leq 90^\circ$

c $\tan\theta = -1$ for $-180^\circ \leq \theta \leq 0^\circ$ d $\tan\theta = 3 - 2\tan\theta$ for $0^\circ \leq \theta \leq 180^\circ$

10 Solve the following equations for the domain $-180^\circ \leq \theta \leq 180^\circ$. Round your answer to two decimal places if necessary.

a $\cos\theta = -0.831$ b $\cos\theta = -\dfrac{1}{2}$ c $\sin\theta = \dfrac{\sqrt{3}}{2}$ d $\sin\theta = 0.238$

e $\tan\theta = 1$ f $\tan\theta = 1.237$

11 Solve the following equations for the domain $360^\circ \leq \theta \leq 720^\circ$. Round your answer to two decimal places if necessary.

a $\sin\theta = 0.3584$ b $\tan\theta = 2.2460$